

# SAI TARRUN PITTA

+1 (657) 751-9260 | [saitarrunpitta@gmail.com](mailto:saitarrunpitta@gmail.com) | <https://linkedin.com/in/saitarrunpitta> | [GitHub](#) | [Portfolio](#)

## SUMMARY

Software engineer with 3+ years building production-scale backend systems and AI-driven applications. Proficient in designing scalable APIs, event-driven services, and cloud-native infrastructure. Specialized in deploying ML models (risk scoring, recommendations, automation) with emphasis on system reliability and code quality.

## EXPERIENCE

### Pacific Life

May 2025 - Dec 2025

Software Engineer Intern

Newport Beach, CA

- Created **TensorFlow risk-scoring inference service** on AWS Lambda and API Gateway, reducing claims triage time from **3 hours to 45 minutes (75% improvement)** and saving **20 hours weekly**.
- Integrated ML inference** into event-driven Kafka pipeline achieving **< 100ms latency for risk updates on 5,000+ weekly claims**; enabled 50 concurrent users without queueing delays.
- Designed and implemented **RAG-enabled document retrieval system** using semantic search and LLM inference over **50K+ internal policy documents**; reduced lookup time by 50%.
- Built **TypeScript API** aggregation layer with **Redis distributed caching**, reducing **dashboard load time** from **4 seconds to 2 seconds (50%)** and API p99 latency from **300ms to 210ms**.
- Built **LangGraph multi-agent system** with human-in-the-loop checkpoints, achieving **98% automation success** and **30% reduction** in failed claims processing.

### Accenture

Jul 2022 - Jun 2024

Software Development Engineer

Hyderabad, India

- Engineered a **gRPC-based wallet service** with distributed locking for a payment platform processing **2M+ daily transactions**, helping maintain **high availability** and reliable **concurrent transaction handling**.
- Optimized **AWS infrastructure** through **EKS right-sizing and Redis caching**, reducing cloud spend by **35% (\$15K/month)**.
- Optimized **PostgreSQL indexes** and query plans to **reduce p95 latency by 18%** across high-throughput transaction services, improving **median query time from 450ms to 180ms**.
- Drove **reliability improvements** by defining SLOs and implementing **Prometheus and Splunk alerting**, reducing **MTTR by 25%** for business-critical services and improving incident triage efficiency.

### Cognizant

May 2021 - Apr 2022

Software Engineer

Hyderabad, India

- Improved **PostgreSQL query performance** and resolved lock contention in Spring Boot services, reducing **API latency by 52% (2.5s to 1.2s)** and **lowering p99 latency by 30%**.
- Integrated **scikit-learn churn prediction model** into retention dashboards, contributing to **12% churn reduction**.
- Built **REST APIs** exposing real-time churn predictions for **5,000+ customers**, enabling proactive retention operations, streamlining outreach workflows, and supporting faster data-driven intervention by customer success teams.
- Implemented **RBAC and input validation**, remediating **40+ high-severity vulnerabilities** for clean security audit.
- Standardized CI/CD with **Docker and GitHub Actions**, reducing release **lead time by 50%** and enabling daily deployments.

## EDUCATION

### California State University, Fullerton

Aug 2024 - May 2026

Master of Science, Computer Science

Fullerton, CA

- Achievements:** Published peer-reviewed IEEE conference paper: "Hardware Trojan Detection with Machine Learning and Power Side-Channels: A Post-Deployment Analysis" (IEEE CNS 2025)
- Coursework:** Advanced Algorithms, Cloud Computing, System Design, Machine Learning, Artificial Intelligence

## PROJECTS

### LLM-Powered Knowledge Retrieval Platform

- Developed a RAG-based application using LLMs, semantic search, and vector retrieval, improving answer relevance by 25% and reducing manual search time by 40%.
- Led optimization of retrieval latency through document chunking and vector database indexing, cutting response times by 30% for domain-specific Q&A.

### Event-Driven E-Commerce Modernization (Monolith to Microservices)

- Developed a personal e-commerce modernization project that migrated a Django monolith to FastAPI microservices using the Strangler pattern, demonstrating zero-downtime migration, improved modularity, and scalable service design.

## SKILLS

- Programming Languages:** Python, Java, JavaScript, SQL, C++, TypeScript
- Frontend Development:** React, Next.js, HTML5/CSS3, Tailwind CSS, Material UI
- Backend & APIs:** Spring Boot, Node.js, FastAPI, Django, gRPC, GraphQL, REST APIs, Microservices
- Data & Caching:** Redis, PostgreSQL, MongoDB, DynamoDB, Elasticsearch, Kafka, RabbitMQ
- Cloud & Infrastructure:** AWS (EC2, Lambda, RDS, S3, DynamoDB, API Gateway), GCP, Azure, Docker, Kubernetes
- DevOps & Observability:** Terraform, Jenkins, GitHub Actions, Prometheus, Grafana, Splunk
- Machine Learning & AI:** TensorFlow, PyTorch, Langchain, LLMs (GPT-4, Claude), RAG Systems, Feature Engineering
- Testing & Quality Assurance:** Pytest, Selenium, Performance Testing, Performance Optimization, Postman, Debugging
- Compliance and Security:** SOC 2, Role-Based Access Control (RBAC), ISO 27001, OAuth 2.0, JWT, OWASP